Shivam Randive

Integrated M.Sc. Physics, SVNIT Surat, India | Machine learning for materials discovery

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Research Profile

I use machine learning and first-principles calculations to discover new Heusler thermoelectrics, materials that convert a temperature difference directly into electrical power. A good thermoelectric needs a high Seebeck coefficient and a low lattice thermal conductivity, and my work addresses both. First, I screened over 100,000 Heusler structures with ML and confirmed the best with first-principles transport calculations, finding 80 promising candidates, 61 never studied experimentally [P1]. Second, because measured thermal conductivity is scarce and calculations overestimate it, I built a dataset of experimental values from the literature and trained a model to predict the measured value, which I then applied to 10 unmeasured half-Heusler compounds [P2]. At NIMS, I am extending this search to unexplored alkali- and alkaline-earth Heusler chemistries with pretrained interatomic potentials. For my PhD, I want to build machine learning that finds new materials reliably.

Education

Integrated M.Sc. (5-year BS–MS) in Physics — **2022 – 2027 (expected)**
S. V. National Institute of Technology (SVNIT) — *Surat, India*
CGPA **8.30 / 10** (top 20% of class)
Thesis: ML for materials discovery and property prediction in Heusler thermoelectrics
Coursework: Quantum Physics I–II, Solid State and Condensed Matter Physics, Density Functional Theory, Computational Physics; Stanford CS224W Machine Learning with Graphs (online)

Semester Exchange Student (selected) — **Upcoming (Jan 2027)**
Indian Institute of Technology (IIT) Delhi — *New Delhi, India*

Publications

First-author manuscripts, both submitted to journals — * corresponding author

[P1] **S. Randive**, R. Dixit, H. Pandey. **High-throughput screening of Heusler alloys for thermoelectrics combining machine-learned densities of states and first-principles transport.** Submitted (2026). Preprint: arXiv:2610.18179.

[P2] **S. Randive***, O. Randive, H. Lungaria, H. Pandey*. **Predicting the experimental lattice thermal conductivity of half-Heusler compounds by machine learning and transfer-function calibration.** Submitted (2026); arXiv preprint forthcoming. Code & data: doi:10.5281/zenodo.23333947

Research Experience

Research Intern · Spin Theory Group, Prof. Terumasa Tadano — **Sep – Dec 2026**
National Institute for Materials Science (NIMS) — *Tsukuba, Japan*

New Heusler thermoelectrics with universal interatomic potentials — ongoing

- *Idea:* build new alkali- and alkaline-earth Heusler compositions combinatorially and rank them by predicted Seebeck coefficient and lattice thermal conductivity, using universal machine-learned interatomic potentials (MLIPs: eSEN, EquiformerV3) pretrained on millions of DFT calculations.
- *Approach:* apply frozen transfer learning to these MLIPs on phonon force data, and predict lattice thermal conductivity through a Phonopy and Phono3py pipeline.

Undergraduate Researcher · Dr. Himanshu Pandey — **Jan 2025 – present**
Department of Physics, SVNIT — *Surat, India*

High-throughput ML screening of Heusler thermoelectrics with DFT validation [P1] — 2026

- *Problem:* searches for new thermoelectrics first pick candidates from the density of states (DOS), often predicted by ML, before running costly transport calculations. No one had tested whether this shortcut picks the right materials.
- *What I did:* tested it on all 106,235 Heusler structures in the DXMag database. ALIGNN graph neural networks predicted the DOS where none existed, and I checked the shortlist with first-principles transport (Quantum ESPRESSO + BoltzTraP2) run through my own automated workflow.
- *Result:* the shortcut is good at choosing what to calculate, needing about **4× fewer expensive calculations** per good material, and it found **80 promising candidates**, 61 never measured. But ML spectra often miss whether a material has a band gap, even when accuracy scores look excellent, so a full calculation must confirm it.

Calibrating ML predictions of lattice thermal conductivity against experiment [P2] 2026

- *Problem:* a good thermoelectric must conduct heat poorly. ML models of lattice thermal conductivity learn from first-principles calculations, but I found that these overestimate the measured value, typically by **about 70%**.
- *What I did:* built one dataset from databases and published papers that records where every value came from (multimodal language models helped read the papers; every value was checked against physics), and a two-parameter correction that turns a calculated value into the one an experiment would measure.
- *Result:* with the correction, ML predicts measured values with **about 30% less error** than standard approaches, close to how much two laboratories disagree. I then predicted the measured value for **10 half-Heusler compounds** not yet measured, for experiments to test.

Descriptor-based ML for perovskites and half-Heuslers [report] Jan – Apr 2025

- Support-vector classifier for perovskite formability (≈90% accuracy); reproduced a Gaussian-process model of lattice constants for 137 half-Heusler compounds.

Next: Phono3py thermal conductivity for the [P1] candidates; crystal generation with MatterGen.

Summer Research Fellow (IASc SRFP) · Prof. Tanusri Saha-Dasgupta **Jun – Jul 2025**
S. N. Bose National Centre for Basic Sciences *Kolkata, India*

Out-of-distribution generalization of graph neural networks for perovskites [report]

- Built a full ALIGNN pipeline (crystal structure to graph to training and inference) and compared a broadly pretrained model with "specialists" trained only on narrow perovskite families.
- *Result:* in metal/non-metal classification, specialists reached >92% accuracy on familiar families but only ≈48% on new double perovskites, while the pretrained model kept ≈80%: breadth of training data matters more than narrow accuracy.

Summer Research Intern **Jun – Jul 2024**
Indian Institute of Technology (IIT) Bhilai *Bhilai, India*

Strain in silicon thin films for stretchable electronics [report]

- COMSOL finite-element model of a silicon film on flexible polyimide; its strain maps fed a collaborator's DFT study of how strain tunes the film's electronic properties.

Open Data & Code

- **heusler-dos-te-screening** [P1] *(lead developer)*: the full screening pipeline (ML models, ranking tools, automated DFT transport workflow) with all 3,625 results and the 80 candidates.
- **Predicted DOS dataset** [P1] *(lead author; Zenodo, release pending)*: ML-predicted DOS for **54,978 Heusler structures** with no DFT spectrum, plus the 30 trained models.
- **half-heusler-kappa-calibration** [P2] *(lead author, archived on Zenodo)*: the 9,495-value thermal-conductivity dataset with the source of every value, the fits, the 10 predictions and the code that rebuilds every figure and number in the paper.

Fellowships & Awards

Fully funded research internship, National Institute for Materials Science (NIMS), Japan 2026
Summer Research Fellowship (SRFP), Indian Academy of Sciences (national, merit-based) 2025
Best Structural Stability and Flight Award (3rd overall), national rocketry, IIT Mandi 2025
Xpecto tech fest; designed and launched a 250 g rocket with my team

Posters & Mentoring

Poster: *AI-driven discovery of Heusler alloys using generative models and electronic structure learning*, National Science Day, SVNIT Surat Feb 2026
Mentoring: two junior students (O. Randive, H. Lungaria), now co-authors on [P2]

Technical Skills

ML methods	Graph neural networks (ALIGNN) and universal MLIPs (eSEN, EquiformerV3): training, fine-tuning, frozen transfer learning; conformal prediction; SHAP
Classical ML	CatBoost, LightGBM, Gaussian processes, support-vector machines
ML libraries	PyTorch, DGL, fairchem, scikit-learn, NumPy, pandas, Matplotlib
First principles	VASP, Quantum ESPRESSO (incl. spin–orbit), BoltzTraP2, Phonopy/Phono3py
Materials data	Materials Project, AFLOW, JARVIS, OQMD, Starrydata; LLM literature mining
Computing	Python, Linux, SLURM GPU clusters, Git, MATLAB, COMSOL Multiphysics, LaTeX